

MATH 473/MTH 573 Assignment # 6

Due on December 9, 2025 (Tuesday)

Students in MTH 473 may skip the questions marked in red.

1. Answer each question concisely.

(a) State sufficient conditions for convergence of Jacobi and Gauss–Seidel methods. Explain why Gauss–Seidel often converges faster.

(b) Explain the main idea behind GMRES.

(c) For which matrices is the Conjugate Gradient (CG) method applicable? Explain the main idea of CG.

2. Implement the following methods from scratch in MATLAB or Python:

(a) Jacobi

(b) Gauss–Seidel

(c) GMRES with Arnoldi iteration

(d) Conjugate Gradient

Your implementation should:

- Take a sparse matrix A , a right-hand side vector $\vec{b}$, and a tolerance as inputs. Jacobi and Gauss–Seidel should additionally accept an initial guess $\vec{x}_0$ and a maximum number of iterations.
- Return the final approximate solution, the number of iterations taken, and the total runtime (in seconds). (Use `tic/toc` in MATLAB or `time.time()` in Python.)

Do not call built-in solvers such as `pcg`, `gmres`, or `bicg`.

3. Consider the linear system $A\vec{x} = \vec{b}$, where $A \in \mathbb{R}^{n \times n}$ is the finite difference discretization of the 1D Poisson equation with Dirichlet boundary conditions:

$$A = (a_{ij}), \quad a_{jj} = 2, \quad a_{j,j-1} = -1, \quad a_{j,j+1} = -1, \quad a_{ij} = 0 \text{ otherwise,}$$

and the right-hand side is

$$\vec{b} = (b_i), \quad b_i = 1, \quad 1 \leq i \leq n.$$

Using your implementations from Question 2, solve the linear system with:

$$n = 200, \quad \vec{x}_0 = \vec{0}, \quad \text{tolerance} = 10^{-3}.$$

Compute and compare the numbers of iterations, the total runtime, and the error

$$\|\vec{x}_{approx} - \vec{x}_{exact}\|_2$$

for Jacobi, Gauss–Seidel, GMRES, and Conjugate Gradient. The exact solution $\vec{x}_{exact}$ should be obtained using the built-in solver (`\` in MATLAB or `solve()` in Python).

(Do NOT print or submit the approximate solution vectors. Only show the comparison table.)